

DPP NO.-8

- Q.1 Number of different words that can be formed using all the letters of the word "DEEPMALA" if two vowels are together and the other two are also together but separated from the first two is :
(A) 960 (B) 1200 (C) 2160 (D) 1440
- Q.2 The number of ways in which 10 boys can take positions about a round table if two particular boys must not be seated side by side is :
(A) $10(9)!$ (B) $9(8)!$ (C) $7(8)!$ (D) none
- Q.3 In a unique hockey series between India & Pakistan, they decide to play on till a team wins 5 matches . The number of ways in which the series can be won by India, if no match ends in a draw is :
(A) 126 (B) 252 (C) 225 (D) none
- Q.4 Number of cyphers at the end of $^{2002}C_{1001}$ is
(A) 0 (B) 1 (C) 2 (D) 200
- Q.5 Three vertices of a convex n sided polygon are selected. If the number of triangles that can be constructed such that none of the sides of the triangle is also the side of the polygon is 30, then the polygon is a
(A) Heptagon (B) Octagon (C) Nonagon (D) Decagon
- Q.6 A gentleman invites a party of $m + n$ ($m \neq n$) friends to a dinner & places m at one table T_1 and n at another table T_2 , the table being round . If not all people shall have the same neighbour in any two arrangement, then the number of ways in which he can arrange the guests, is
(A) $\frac{(m+n)!}{4mn}$ (B) $\frac{1}{2} \frac{(m+n)!}{mn}$ (C) $2 \frac{(m+n)!}{mn}$ (D) none
- Q.7 There are 12 guests at a dinner party . Supposing that the master and mistress of the house have fixed seats opposite one another, and that there are two specified guests who must always, be placed next to one another ; the number of ways in which the company can be placed, is:
(A) $20 \cdot 10!$ (B) $22 \cdot 10!$ (C) $44 \cdot 10!$ (D) none
- Q.8 Let P_n denotes the number of ways of selecting 3 people out of ' n ' sitting in a row , if no two of them are consecutive and Q_n is the corresponding figure when they are in a circle . If $P_n - Q_n = 6$, then ' n ' is equal to :
(A) 8 (B) 9 (C) 10 (D) 12
- Q.9 Define a 'good word' as a sequence of letters that consists only of the letters A, B and C and in which A never immediately followed by B, B is never immediately followed by C, and C is never immediately followed by A. If the number of n -letter good words are 384, find the value of n .
- Q.10 Six married couple are sitting in a room. Find the number of ways in which 4 people can be selected so that
(a) they do not form a couple (b) they form exactly one couple
(c) they form at least one couple (d) they form atmost one couple

- Q.11 Fifty college teachers are surveyed as to their possession of colour TV, VCR and tape recorder. Of them, 22 own colour TV, 15 own VCR and 14 own tape recorders. Nine of these college teachers own exactly two items out of colour TV, VCR and tape recorders ; and, one college teacher owns all three. how many of the 50 college teachers own none of three, colour TV, VCR or tape recorder?
 (A) 4 (B) 9 (C) 10 (D) 11
- Q.12 There are counters available in x different colours. The counters are all alike except for the colour. The total number of arrangements consisting of y counters, assuming sufficient number of counters of each colour, if no arrangement consists of all counters of the same colour is :
 (A) $x^y - x$ (B) $x^y - y$ (C) $y^x - x$ (D) $y^x - y$
- Q.13 There are $(p + q)$ different books on different topics in Mathematics. ($p \neq q$)
 If L = The number of ways in which these books are distributed between two students X and Y such that X get p books and Y gets q books.
 M = The number of ways in which these books are distributed between two students X and Y such that one of them gets p books and another gets q books.
 N = The number of ways in which these books are divided into two groups of p books and q books then,
 (A) $L = M = N$ (B) $L = 2M = 2N$ (C) $2L = M = 2N$ (D) $L = M = 2N$

The question given below contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE is correct. Choose the correct alternative.

- Q.14 Statement 1: The sum ${}^{40}C_0 \cdot {}^{60}C_{10} + {}^{40}C_1 \cdot {}^{60}C_9 + \dots + {}^{40}C_{10} \cdot {}^{60}C_0$ equals ${}^{100}C_{10}$.
because
 Statement 2: Number of ways of selecting 10 students out of 40 boys and 60 girls is ${}^{100}C_{10}$.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

MATCH THE COLUMN

- | Q.15 | Column-I | Column-II |
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| (A) | In a plane a set of 8 parallel lines intersect a set of n parallel lines, that goes in another direction, forming a total of 1260 parallelograms. The value of n is equal to | (P) 6 |
| (B) | If $\frac{{}^{n-1}P_3}{{}^nP_4} = \frac{1}{9}$ then n is equal to | (Q) 9 |
| (C) | Number of ways in which 5 persons A, B, C, D and E can be seated on round table if A and D do not sit next to each other | (R) 10 |
| (D) | Number of cyphers at the end of the number ${}^{50}P_{25}$ | (S) 12 |